Analysis on Hate Crime Rates Across the U.S. (Part 2)

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GitHub Repository: https://github.com/ChloeSong2468/Study-on-Hate-Crime-Rates-across-United-States

1 Introduction

Hate crimes are a serious concern across the United States, and understanding what socioeconomic factors are linked to them can help inform our policies. This project looks into whether states with higher median household incomes have different hate crime rates than states with lower median household incomes. The research question is:

"Is the distribution of hate crime rates different between states with above-median household income and states with below-median household income?"

In Part 1, the EDA showed that the hate crime rate is right-skewed across the states, with some high rate outliers pulling the mean above the median. Somewhat surprisingly, the correlation between median household income and hate crime rate was weakly positive ($r = 0.351$), suggesting that states with higher income do not tend to have lower hate crime rates. This report tests whether that difference is statistically significant.

2 Methods

2.1 Test Selection

To answer the research question, we use a Wilcoxon rank-sum test. States are split into two groups based on whether their median household income is above or below the median income across all states. The test compares the distribution of hate crime rates between these two groups.

```
##
## Above Median Below Median
##           24           23
```

The Wilcoxon rank-sum test is appropriate here because: 1. We are comparing a continuous outcome (hate crime rate) across two independent groups (above vs. below median income). 2. The two groups are independent; a state's income group has no influence on another state's. 3. The test assesses whether one group tends to have systematically higher or lower hate crime rates than the other.

2.2 Hypotheses

F_1 = distribution of hate crime rates for above-median income states F_2 = distribution of hate crime rates for below-median income states

$$H_0 : F_1 = F_2$$
$$H_a : F_1 \neq F_2$$

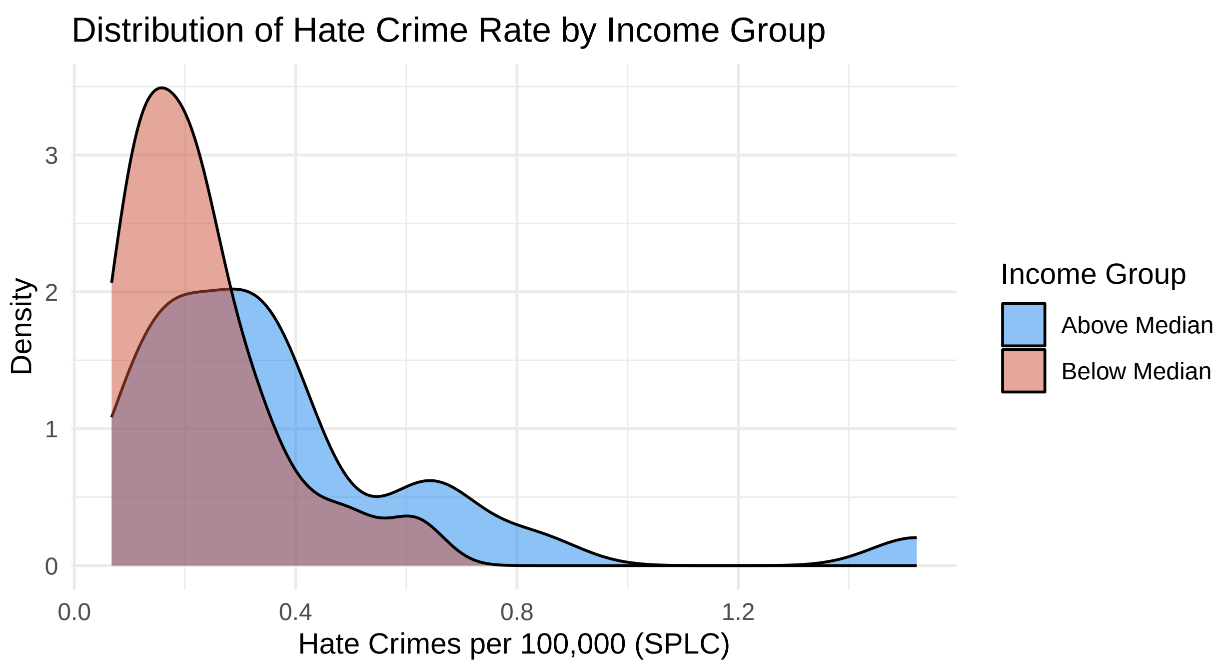
We use a significance level of $\alpha = 0.05$.

2.3 Assumption Verification

1. Independence: Each observation is a U.S. state. States are independent units; the hate crime rate in one state does not directly influence the rate in another.

2. Continuity: The hate crime rate is a continuous variable, satisfying this requirement for the Wilcoxon rank-sum test.

3. Similar shape: The Wilcoxon rank-sum test assumes the two distributions have a roughly similar shape. We verify this with density plots.



Both distributions appear right-skewed with similar shapes, satisfying the assumption.

3 Results

```
#wilcoxon rank-sum test (two-sided)
wilcox.test(hate_crimes_per_100k_splc ~ income_group, data = hc)
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  hate_crimes_per_100k_splc by income_group
## W = 380, p-value = 0.02665
## alternative hypothesis: true location shift is not equal to 0
```

```
## # A tibble: 2 x 4
##   income_group     n median    sd
##   <chr>        <int>  <dbl> <dbl>
## 1 Above Median    24  0.323 0.313
## 2 Below Median    23  0.2   0.137
```

The Wilcoxon rank-sum test produced a p-value of 0.027, which is less than $\alpha = 0.05$, so the result is statistically significant. States in the above-median income group tended to have higher hate crime rates than states in the below-median income group. The test statistic W reflects the number of times a hate crime rate from the above-median group ranked higher than one from the below-median group across all pairwise comparisons.

4 Conclusions

4.1 Interpretation

Since the p-value (0.027) is less than $\alpha = 0.05$, we reject the null hypothesis. There is sufficient evidence to conclude that the distribution of hate crime rates differs significantly between states with above and below median household incomes. Interestingly, above-median income states tend to have higher hate crime rates than below-median income states; opposite of what we might expect. This likely reflects the under-reporting issue noted earlier, where wealthier states may have stronger infrastructure for reporting hate crimes rather than genuinely experiencing more incidents.

4.2 Limitations

Some limitations are worth noting. First, hate crime statistics are subject to significant under-reporting, and reporting rates likely vary systematically by state wealth; this could artificially inflate rates in wealthier states and confound our results. Second, splitting states into just two income groups discards information about the continuous nature of income.

4.3 Generalizability

Since the dataset includes 46 U.S. states and D.C. (minus 4 states with missing SPLC data), these findings describe the U.S. states included in the data as of the mid-2010s rather than a random sample. The 4 missing states are excluded because SPLC did not record hate crime data for them during that time, not because of specific selections that have to do with their income levels or hate crime rates. Their absence is unlikely to meaningfully bias the conclusions.

That said, the results should not be generalized beyond the states included since the data are not based on random sampling. Still, the findings could give insight into how socioeconomic conditions relate to hate crime reporting across big political units, and motivate further investigation using individual or county-level data.

5 References

Majumder, M. (2017, January 23). *Higher rates of hate crimes are tied to income inequality.* FiveThirtyEight. https://fivethirtyeight.com/features/higher-rates-of-hate-crimes-are-tied-to-income-inequality/

FiveThirtyEight. (2017). *Hate crimes data* [Data set]. GitHub. https://github.com/fivethirtyeight/data/blob/master/hate-crimes/hate_crimes.csv

Federal Bureau of Investigation. (2016). *Hate crime statistics, 2015.* U.S. Department of Justice. https://ucr.fbi.gov/hate-crime/2015

Southern Poverty Law Center. (2016). *Update: Hate incidents in the days following the election.* https://www.splcenter.org/resources/hatewatch/update-incidents-hateful-harassment-election-day-now-number-701

United States Census Bureau. (2016). *American Community Survey 5-year estimates.* https://www.census.gov/library/visualizations/interactive/acs-5year-datamap.html

Gemini. (2026, April 19). Response to: “if i excluded 4 states out of 51 (including DC) because of missing data, should i not generalize the findings beyond the states that were included (since the data are not based on random sampling)?” https://gemini.google.com/